

Program 1:

Dockerizing flask application using custom dockerfile

Step1 : create the [app.py](#) ,requirements.txt and Dockerfile inside the folder

```
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/prg1$ cat app.py
from flask import Flask

app = Flask(__name__)

app.route("/")
def hello():
    return "Hello world!"

if __name__=="__main__":
    app.run(host="0.0.0.0",port=5000)
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/prg1$ cat Dockerfile
FROM python:3.9-slim

WORKDIR /app

COPY requirements.txt .

RUN pip install --no-cache-dir -r requirements.txt

COPY . .

ENV FLASK_APP=app.py
ENV FLASK_RUN_HOST=0.0.0.0
ENV FLASK_RUN_PORT=5000

EXPOSE 5000

CMD ["flask","run"]

1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/prg1$ cat requirements.txt
flask
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/prg1$ █
```

Step 2: build the container

docker build -t program-1 .

Step 3: run the container

```
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530: ~/Dockerfiles/program_1
program-1    latest    393ccc1b89e2    2 minutes ago    132MB
<none>      <none>    a662902dd135    11 hours ago    132MB
secondfile  latest    06fba82a2a27    2 days ago      17.8MB
wordpress   latest    7332768c717f    4 weeks ago     734MB
hello-world latest    1b44b5a3e06a    2 months ago    10.1kB
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/program_1$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS    NAMES
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/program_1$ docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS    NAMES
239ca8fdd0fc   program-1    "flask run"              24 seconds ago    Exited (2) 23 seconds ago    prg1_flask-container
561e3d15ea4f   a662902dd135    "flask run"              11 hours ago     Exited (1) 11 hours ago     flask-container1
361842c0aeba   a662902dd135    "flask run"              11 hours ago     Exited (1) 11 hours ago     flask-container
f3e27848d382   wordpress    "docker-entrypoint.s..." 2 days ago       Exited (0) 2 days ago       container1
42aeba89be41   secondfile    "redis-server"           2 days ago       Exited (0) 2 days ago       zen_ritchie
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/program_1$ ls
app.py  Dockerfile  requirements.txt
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/program_1$ cat Dockerfile
#This is the dockerfile which is a set of instruction to create images

FROM python:3.9-slim

WORKDIR /app

#copy and install the requirements

COPY requirements.txt .

RUN pip install --no-cache-dir -r requirements.txt

COPY . .

ENV FLASK_APP=app.py
ENV FLASK_RUN_HOST=0.0.0
ENV FLASK_RUN_PORT = 5000

EXPOSE 5000

CMD ["flask","run"]

1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/program_1$ docker run -d -p 5000:5000 --name flask-container1 program-1
docker: Error response from daemon: Conflict. The container name "/flask-container1" is already in use by container "561e3d15ea4f4a627f10e8beab4db54dcc6e38a6ba9a03bfd874b49804c520ba".
You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/program_1$ docker ps -a
```